A 46 μW 13 b 6.4 MS/s SAR ADC with Background Mismatch and Offset Calibration

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A 46 μW 13 b 6.4 MS/s SAR ADC With Background Mismatch and Offset Calibration

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***Abstract*—A 6.4 MS/s 13 b ADC with a low-power background calibration for DAC mismatch and comparator offset errors is presented. Redundancy deals with DAC settling and facilitates calibration. A two-mode comparator and 0.3 fF capacitors reduce power and area. The background calibration can directly detect the sign of the dynamic comparator offset error and the DAC mismatch errors and correct both of them simultaneously in a stepwise feedback loop. The calibration achieves 20 dB spur reduction with little area and power overhead. The chip is implemented in 40 nm CMOS and consumes 46 μW from a 1 V supply, and achieves 64.1 dB SNDR and a FoM of 5.5 fJ/conversion-step at Nyquist.**

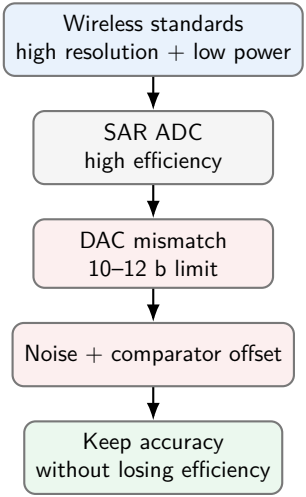
Introduction: Motivation and Challenges

Context

- Wireless standards such as **IEEE 802.15.4g** need **high-resolution** ADCs at MS/s rates with **very low power**.
- SAR ADCs are attractive because they already offer excellent **energy efficiency**.

Main problems above 10 bits

- **DAC mismatch**: intrinsic capacitor matching in modern CMOS is practically limited to about **10–12 bits**.
- **Noise and comparator offset**: these begin to limit usable resolution.
- Larger capacitors help matching, but increase **power** and **area**, and can reduce **speed**.



Introduction: Prior Fixes and This Paper's Solution

Why prior fixes are costly

- **DSP / LMS** and **statistics-based calibration** add digital processing, sample collection, or foreground time → more **area**, **power**, and **latency**.
- **Two-stage pipelined SAR + amplifier** can relax comparator noise, but the amplifier is hard to design and introduces new analog errors.
- **Comparator oversampling** improves reliability, but extra comparisons reduce **speed**.

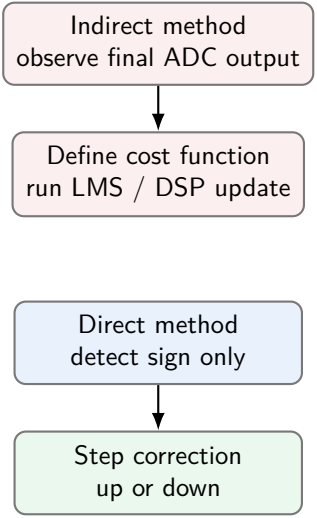
This paper proposes

- A **13-bit SAR ADC** with a **reconfigurable comparator**.
- **Low-power on-chip background calibration** for **DAC mismatch** and **dynamic comparator offset**.

Reported design summary	
Resolution	13 b
Sampling rate	6.4 MS/s
Power	46 μW
SNDR	64.1 dB
ENOB	10.4 b
FoM	5.5 fJ/conv-step

Section II: Design Philosophy

- **Background** calibration can keep running during normal ADC operation; foreground methods usually need special conditions.
- Prior work often detects errors **indirectly**: analyze final ADC output and adapt coefficients with DSP/LMS-like logic.
- This paper detects errors **directly**: determine only the **sign** of the error, then step correction in the opposite direction.
- Intuition: a **hardware-efficient sign-gradient update** with much less arithmetic.



Section II: ADC Architecture (Fig. 2)

- The S&H samples the **differential input** on the capacitive DAC array.
- Comparator + SAR logic + **monotonic switching** perform the successive-approximation search.
- The ADC uses **15 cycles** for a **13-bit** conversion, with **2 redundant cycles**.
- The **7th** redundant cycle helps DAC settling / mismatch calibration; the **11th** helps tolerate comparator noise / offset.
- A digital adder maps the **15-bit raw code** to the final **13-bit output**; an optional **16th cycle** is used for calibration.

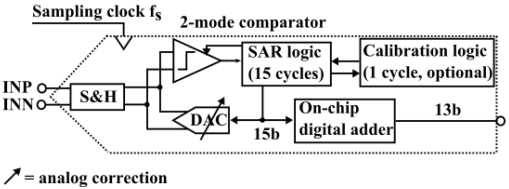


Fig. 2. Block diagram of a 13 b SAR ADC.

Section II: Sign-Based Calibration Loops

Comparator offset detection

- Add the **16th cycle**; keep the DAC code the same as in cycle 15.
- Switch the comparator from fine mode back to coarse mode.
- Compare D_{15} and D_{16} to get the **sign** of the offset error.

DAC mismatch detection

- Redundancy means two different internal DAC codes can represent the same final output.
- If code A is observed, the controller enables the 16th cycle and forces the alternate code B.
- Comparing cycle 15 under A and cycle 16 under B reveals the **sign** of the mismatch.

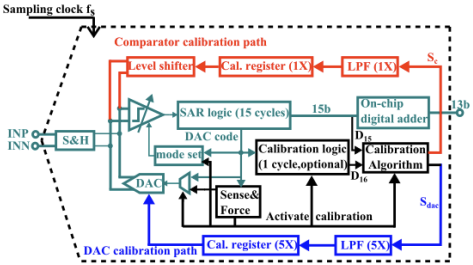
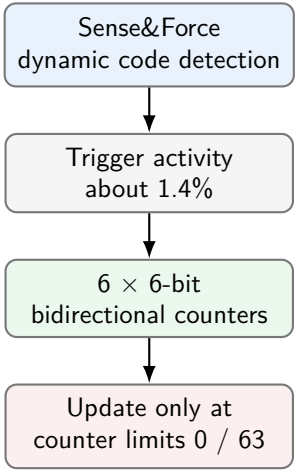


Fig. 10. Block diagram of the implemented 13 b SAR ADC.

Section III: Low-Power Calibration Logic

How they made the controller cheap

- **Dynamic Sense&Force**: code-detection logic mostly stays reset/inactive unless a calibration trigger appears.
- **Low activation rate**: specific X- and Y-code patterns reduce calibration activity to about **1.4%** of conversions.
- **Digital LPF**: six **6-bit bidirectional counters** are used, one comparator path and five DAC paths.
- Counters move **up/down** based on the error sign and only update the calibration register at $(0, 63)$.



Section III: Analog Correction Hardware and Measured Impact

Analog correction hardware

- **DAC calibration**: custom **75 aF** capacitors provide **0.25 LSB** correction steps for the first five DAC capacitors.
- **Comparator correction**: programmable capacitor banks C_a and C_b create a compensating input step.
- Comparator correction range is up to ± 63 LSB with **1 LSB** steps.

Takeaway

- Tiny analog correction elements plus sparse digital control let the calibration overhead stay very small.

Implementation summary	
Process	40 nm CMOS
Logic type	Dynamic logic
LPF counters	6-bit bidirectional
DAC step	0.25 LSB
DAC trim cap	75 aF
Comp. step	1 LSB
Digital cal area	0.0017 mm^2

Measured impact
20 dB spur reduction
64.1 dB SNDR
400k-cycle settling

Measurements Summary

Measured / Reported Results	
Process	40 nm CMOS
Supply voltage	1 V
Active area	0.0675 mm^2
Digital calibration area	0.0017 mm^2
Analog correction area	0.0009 mm^2
Sampling rate	6.4 MS/s
Resolution	13 b
Total power	46 μW
Calibration power overhead	$\sim$5%
ENOB	10.4 b
SNDR	64.1 dB
SFDR	81.9 dB
FoM	5.5 fJ/conv-step
Spur reduction from calibration	$\sim$20 dB
Convergence time	$\sim$400k cycles